

Theorem 2.27 has two corollaries.

Corollary 2.28 [4] *For every connected bridgeless graph G with radius r ,*

$$rc(G) \leq r(r+2).$$

Moreover, for every integer $r \geq 1$, there exists a bridgeless graph with radius r and $rc(G) = r(r+2)$.

Corollary 2.29 [4] *For every connected bridgeless graph G with radius r and chordality k ,*

$$rc(G) \leq \sum_{i=1}^r \min\{2i+1, k\} \leq rk.$$

Moreover, for every two integers $r \geq 1$ and $3 \leq k \leq 2r+1$, there exists a bridgeless graph G with radius r and chordality k such that $rc(G) = \sum_{i=1}^r \min\{2i+1, k\}$.

Corollary 2.28 answered the above question in the affirmative, the bound is sharp and is a function of the radius r alone. Basavaraju, Chandran, Rajendraprasad and Ramaswamy also demonstrated that the bound cannot be improved even if we assume stronger connectivity by constructing a κ -vertex-connected graph of radius r whose rainbow connection number is $r(r+2)$ for any two given integers $\kappa, r \geq 1$: Let $s(0) := 0, s(i) := 2\sum_{j=r}^{r-i+1} j$ for $1 \leq i \leq r$ and $t := s(r) = r(r+1)$. Let $V = V_0 \uplus V_1 \uplus \dots \uplus V_t$ where $V_i = \{x_{i,0}, x_{i,1}, \dots, x_{i,\kappa-1}\}$ for $0 \leq i \leq t-1$ and $V_t = \{x_{t,0}\}$. Construct a graph $X_{r,\kappa}$ on V by adding the following edges. $E(X) = \{\{x_{i,j}, x_{i',j'}\} : |i-i'| \leq 1\} \cup \{\{x_{s(i),0}, x_{s(i+1),0}\} : 0 \leq i \leq r-1\}$.

Corollary 2.29 generalises a result from [11] that the rainbow connection number of any bridgeless chordal graph is at most three times its radius as the chordality of a chordal graph is three.

In [9], Caro, Lev, Roditty, Tuza and Yuster also derived a result which gives an upper bound for rainbow connection number according to the order and the number of vertex-disjoint cycles. Here $\chi'(G)$ is the chromatic index of G .

Theorem 2.30 [9] *Suppose G is a connected graph with n vertices, and assume that there is a set of vertex-disjoint cycles that cover all but s vertices of G . Then $rc(G) < 3n/4 + s/4 - 1/2$. In particular:*

- (i). *If G has a 2-factor then $rc(G) < 3n/4$.*
- (ii). *If G is k -regular and k is even then $rc(G) < 3n/4$.*
- (iii). *If G is k -regular and $\chi'(G) = k$ then $rc(G) < 3n/4$.*

Another approach for achieving upper bounds is based on the size (number of edges) m of the graph. Those type of sufficient conditions are known as Erdős-Gallai type results. Research on the following Erdős-Gallai type problem has been started in [34].

Problem 2.31 [34] *For every k , $1 \leq k \leq n-1$, compute and minimize the function $f(n, k)$ with the following property: If $|E(G)| \geq f(n, k)$, then $rc(G) \leq k$.*

In [34], Kemnitz and Schiermeyer gave a lower bound for $f(n, k)$, i.e., $f(n, k) \geq \binom{n-k+1}{2} + (k-1)$. They also computed $f(n, k)$ for $k \in \{1, n-2, n-1\}$, i.e., $f(n, 1) = \binom{n}{2}$, $f(n, n-1) = n-1$, $f(n, n-2) = n$, and obtained $f(n, 2) = \binom{n-1}{2} + 1$ for $k = 2$.

In [48], Li and Sun provided a new approach to investigate the rainbow connection number of a graph G according to some constraints to its complement graph $\overline{G}$. They gave two sufficient conditions to guarantee that $rc(G)$ is bounded by a constant. By using the fact that $rc(G) \leq rc(H)$ where H is a connected spanning subgraph of a connected graph G , and the structure of its complement graph as well as Propositions 2.4 and 2.5, they derived the following result.

Theorem 2.32 [48] *For a connected graph G , if $\overline{G}$ does not belong to the following two cases: (i) $\text{diam}(\overline{G}) = 2, 3$, (ii) $\overline{G}$ contains exactly two connected components and one of them is trivial, then $rc(G) \leq 4$. Furthermore, this bound is best possible.*

For the remaining cases, $rc(G)$ can be very large as discussed in [48]. So They add a constraint: let $\overline{G}$ be triangle-free, then G is claw-free. And they derived the following result. In their argument, Theorem 2.40 is useful.

Theorem 2.33 [48] *For a connected graph G , if $\overline{G}$ is triangle-free, then $rc(G) \leq 6$.*

The readers may consider the rainbow connection number of a graph G according to some other condition to its complement graph.

Chen, Li and Lian [17] investigated Nordhaus-Gaddum-type result. A Nordhaus-Gaddum-type result is a (sharp) lower or upper bound on the sum or product of the values of a parameter for a graph and its complement. The name “Nordhaus-Gaddum-type” is so given because it is Nordhaus and Gaddum [49] who first established the following type of inequalities for chromatic number of graphs in 1956.

Theorem 2.34 [17] *Let G and $\overline{G}$ be connected with $n \geq 4$, then*

$$4 \leq rc(G) + rc(\overline{G}) \leq n + 2.$$

Furthermore, the upper bound is sharp for $n \geq 4$ and the low bound is sharp for $n \geq 8$.

They also proved that $rc(G) + rc(\overline{G}) \geq 6$ for $n = 4, 5$; and $rc(G) + rc(\overline{G}) \geq 5$ for $n = 6, 7$ and these bounds are best possible.

2.3 For some graph classes

Some graph classes, such as line graphs, have many special properties, and by these properties we can get some interesting results on their rainbow connection numbers in terms of some graph parameters. For example, in [9] Caro, Lev, Roditty, Tuza and Yuster derived Theorem 2.23 according to the ear-decomposition of a 2-connected graph. In this subsection, we will introduce some results on rainbow connection numbers of line graphs, etc.

In [42] and [43], Li and Sun studied the rainbow connection numbers of line graphs in the light of particular properties of line graphs shown in [30] and [31]. They gave two sharp upper bounds for rainbow connection number of a line graph and one sharp upper bound for rainbow connection number of an iterated line graph.

Recall the *line graph* of a graph G is the graph $L(G)$ (or $L^1(G)$) whose vertex set $V(L(G)) = E(G)$, and two vertices e_1, e_2 of $L(G)$ are adjacent if and only if they are adjacent in G . The *iterated line graph* of a graph G , denoted by $L^2(G)$, is the line graph of the graph $L(G)$. More generally, the *k-iterated line graph* $L^k(G)$ is the line graph of $L^{k-1}(G)$ ($k \geq 2$). We also need the following new terminology.

For a connected graph G , we call G a *clique-tree-structure*, if it satisfies the following condition: each block is a maximal clique. We call a graph H a *clique-forest-structure*, if H is a disjoint union of some clique-tree-structures, that is, each component of a clique-forest-structure is a clique-tree-structure. By the above condition, we know that any two maximal cliques of G have at most one common vertex. Furthermore, G is formed by its maximal cliques. The *size* of a clique-tree(forest)-structure is the number of its maximal cliques. An example of clique-forest-structure is shown in Figure 2.2. If each block of a

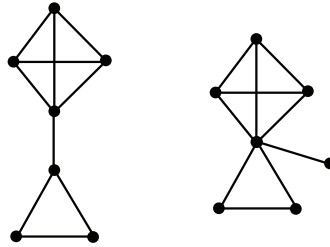


Figure 2.2 A clique-forest-structure with size 6 and 2 components.

clique-tree-structure is a triangle, we call it a *triangle-tree-structure*. Let ℓ be the size of a triangle-tree-structure. Then, by definition, it is easy to show that there are $2\ell + 1$ vertices in it. Similarly, we can give the definition of a *triangle-forest-structure* and there are $2\ell + c$ vertices in a triangle-forest-structure with size ℓ and c components. We denote n_2 the number of inner vertices (degrees at least 2) of a graph.

Theorem 2.35 [43] *For any set $\mathcal{T}$ of t edge-disjoint triangles of a connected graph G , if*